

HELENA, MT, USA

WMO: 727720

Lat: 46.606N

Lon: 111.964W

Elev: 3828

StdP: 12.77

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24144

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -10.6	(c) -5.3	(d) -18.2	(e) 2.3	(f) -9.6	(g) -12.3	(h) 3.3	(i) -4.8	(j) 27.3	(k) 40.3	(l) 24.4	(m) 40.1	(n) 3.6	(o) 280	(p) 0.612

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.7	93.4	61.5	90.1	60.7	86.8	59.8	64.4	85.5	62.8	83.8	61.4	81.8	10.0	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.0	82.8	68.6	55.8	76.4	67.9	53.7	70.6	67.6	31.4	85.3	30.1	83.7	29.0	81.5	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 24.3	(b) 20.6	(c) 18.4	(e) -18.9	(f) 98.7	(g) 8.4	(h) 2.5	(i) -25.0	(j) 100.4	(k) -30.0	(l) 101.9	(m) -34.7	(n) 103.3	(o) -40.8	(p) 105.1		
(a) 24.3	(b) 20.6	(c) 18.4	(e) -18.9	(f) 98.7	(g) 8.4	(h) 2.5	(i) -25.0	(j) 100.4	(k) -30.0	(l) 101.9	(m) -34.7	(n) 103.3	(o) -40.8	(p) 105.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.8	23.4	26.7	35.9	44.5	53.6	61.9	71.3	69.0	59.2	45.8	33.1	23.5	
	DBStd	19.29	13.66	13.01	11.15	7.97	7.75	7.21	5.85	5.79	8.11	8.58	11.40	12.57	
	HDD50	3700	826	652	442	199	50	5	0	0	18	178	509	821	
	HDD65	7466	1290	1071	902	615	361	140	15	30	205	595	956	1286	
	CDD50	2158	0	0	6	33	161	362	660	590	294	48	4	0	
	CDD65	450	0	0	0	0	7	47	210	155	31	0	0	0	
	CDH74	6419	0	0	0	18	195	730	2789	2096	570	21	0	0	
	CDH80	2733	0	0	0	0	42	251	1332	918	187	3	0	0	
Wind	WSAvg	6.6	5.5	6.4	7.3	8.2	7.7	7.7	6.9	6.3	6.1	6.4	5.9	5.5	
Precipitation	PrecAvg	11.4	0.5	0.4	0.6	1.0	1.8	2.0	1.2	1.3	1.0	0.6	0.5	0.5	
	PrecMax	20.9	2.8	1.2	1.6	3.0	6.1	4.6	4.7	4.2	3.4	2.7	1.2	1.5	
	PrecMin	6.3	0.0	0.0	0.0	0.1	0.3	0.1	0.1	0.0	0.1	0.0	0.0	0.0	
	PrecStd	3.0	0.5	0.3	0.4	0.6	1.1	1.1	1.1	0.9	0.8	0.6	0.3	0.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.3	57.8	67.5	76.6	85.2	92.6	98.5	96.2	90.4	76.5	63.8	53.8	
		MCWB	40.4	42.9	47.3	52.4	57.2	62.2	64.0	61.8	59.9	53.7	47.7	41.8	
	2%	DB	46.9	50.9	61.4	70.3	79.6	86.8	94.8	92.3	85.2	70.3	56.9	47.6	
		MCWB	38.0	40.2	44.7	49.7	55.7	61.1	62.2	60.7	57.7	51.3	43.9	38.5	
	5%	DB	43.5	46.6	56.3	64.7	74.5	81.9	91.3	89.1	80.8	64.6	52.2	43.7	
		MCWB	35.8	37.5	42.4	47.3	54.3	59.1	61.2	60.0	56.7	48.8	41.8	36.0	
	10%	DB	40.3	42.7	51.6	59.4	69.6	77.7	88.0	85.5	76.0	59.9	48.2	40.0	
		MCWB	33.8	35.4	40.2	45.0	52.1	57.4	60.4	59.4	55.0	46.7	39.5	33.6	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.9	44.2	48.9	54.1	59.5	65.4	67.2	65.6	62.9	55.3	49.3	42.4
			MCDB	49.9	55.9	65.4	72.5	78.6	84.3	90.1	86.6	80.3	73.1	61.9	52.3
2%		WB	38.5	40.9	45.6	50.7	57.2	62.5	65.0	63.5	60.2	52.5	44.9	39.2	
		MCDB	46.0	49.6	59.5	67.7	75.1	82.4	86.8	84.6	79.5	67.3	55.2	46.6	
5%		WB	36.2	38.2	43.0	48.3	55.3	60.4	63.6	61.9	58.3	50.0	42.4	36.6	
		MCDB	42.8	45.8	55.0	63.2	71.8	78.5	85.4	82.9	76.9	63.2	51.4	43.1	
10%		WB	34.0	35.6	40.6	45.8	53.1	58.6	62.0	60.4	56.0	47.6	40.0	33.9	
		MCDB	39.9	42.2	50.9	58.4	67.5	74.7	82.8	80.8	74.1	58.9	47.7	39.5	
Mean Daily Temperature Range	5% DB	MDBR	17.2	18.3	21.2	22.6	23.2	24.5	29.7	30.1	27.1	22.4	18.4	16.4	
		MCDBR	19.8	23.0	29.1	32.8	32.5	32.5	34.5	36.0	35.8	31.1	24.3	18.9	
		MCWBR	13.1	14.3	15.7	16.4	14.1	13.4	12.4	13.6	15.0	15.6	14.2	12.5	
	5% WB	MCDBR	19.1	22.2	27.7	30.7	29.8	30.2	31.5	32.3	32.3	28.9	22.8	18.7	
		MCWBR	13.1	14.4	15.5	15.6	13.3	13.3	12.5	12.8	14.1	15.0	13.8	12.8	
Clear-Sky Solar Irradiance	taub	0.249	0.260	0.275	0.313	0.321	0.329	0.336	0.351	0.318	0.290	0.262	0.248		
	taud	2.489	2.492	2.554	2.455	2.493	2.503	2.497	2.425	2.497	2.572	2.565	2.509		
	Ebn at Noon	278	297	306	300	300	296	293	284	286	281	271	266		
	Edh at Noon	20	24	27	33	33	33	33	34	29	23	19	17		
All-Sky Solar Radiation	RadAvg	480	790	1171	1520	1765	2051	2271	1913	1402	876	512	372		
	RadStd	41	57	78	94	156	149	128	105	123	75	46	27		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	-0.67	-1.21	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	-1.03	N/A	N/A	N/A	N/A

Nomenclature: See separate page